

Trader Performance vs Market Sentiment

Exploratory Data Analysis Report

Crypto Perpetuals Trading Cohort • 2023–2025

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1. Executive Summary

This report examines whether and how the Crypto Fear & Greed Index (F&G) relates to trader performance across a cohort of 32 wallets trading perpetual contracts between May 2023 and May 2025. Using 211,224 trade records merged with daily F&G scores, we identify significant variation in profitability, position direction, trade sizing, and win rates across sentiment regimes.

Four structural findings define the narrative:

- Sentiment extremes outperform the middle - Extreme Greed produces the highest avg PnL (\$67.9) and win rate (46.5%), while the F&G range 60-67 is a near-dead zone with daily aggregate PnL collapsing to ~\$2.5K.
- Shorts are systematically underused despite outperforming longs in almost every sentiment regime except Greed, yet 63-67% of trades are long-biased during Fear periods.
- The F&G index itself has near-zero correlation with daily PnL ($r = -0.08$), but trade count correlates positively ($r = 0.36$) - volume, not sentiment direction, drives aggregate profitability.
- Trader archetypes are clearly differentiated - a market maker, two mirror-strategy specialists, and several sentiment-conditional traders all coexist profitably, suggesting the cohort is diverse enough to study strategy variation.

2. Dataset Overview

2.1 Fear & Greed Index

The F&G index assigns a daily score from 0 (Extreme Fear) to 100 (Extreme Greed) based on volatility, market momentum, social media, dominance, and trends. The 2,644-day series used in this analysis spans February 2018 to May 2025, though trade data only covers the 2023–2025 window.

Classification	Score range	Days in series	Trades matched	Avg F&G	%
Extreme Fear	0-24	508	21,400	~18	10%
Fear	25-44	781	61,837	~36	29%
Neutral	45-54	396	37,686	~49	18%
Greed	55-74	633	50,303	~65	24%
Extreme Greed	75-100	326	39,992	~83	19%

2.2 Trade data

The historical dataset covers 32 unique trader wallets executing perpetual futures trades on a decentralised platform. Key fields include Execution Price, Size USD, Side (BUY/SELL), Direction (Open/Close Long/Short), Closed PnL, Fee, and Timestamp.

- 246 unique coins traded; top coins by volume: HYPE (68K trades), @107 (30K), BTC (26K), ETH (11K), SOL (10K).
- Median Closed PnL is \$0 across all sentiment regimes - PnL is driven by a small number of high-magnitude trades.
- Activity exploded in late 2024: from ~2K trades/month in mid-2024 to 52K/month by April 2025, driven by multiple new traders entering the cohort.

3. Sentiment-Level Analysis

3.1 PnL and win rate by sentiment regime

Across all 211,224 trades, average Closed PnL and win rate follow a non-linear pattern across sentiment regimes — neither pure fear nor pure greed is uniformly bad or good.

Sentiment	Avg PnL (\$)	Median PnL	Win rate	Avg size (USD)	Trade s
Extreme Greed	\$67.9	\$0	46.5%	\$3,112	39,992
Fear	\$54.3	\$0	42.1%	\$7,816	61,837
Greed	\$42.7	\$0	38.5%	\$5,737	50,303
Neutral	\$34.3	\$0	39.7%	\$4,783	37,686
Extreme Fear	\$34.5	\$0	37.1%	\$5,350	21,400

Key insight

Extreme Greed is the best-performing regime on every metric. Fear outperforms Greed despite lower win rate, because traders are sizing up (avg \$7,816 vs \$5,737) and capturing larger moves. Neutral is the weakest regime — low size, low win rate, low PnL.

3.2 Long vs Short performance by sentiment

The most structurally important finding in the dataset: shorts outperform longs in every sentiment regime except Greed, yet the majority of traders are long-biased during Fear and Extreme Fear periods.

Sentiment	Avg Long PnL	Avg Short PnL	Short edge	Long ratio
Fear	\$41.4	\$95.2	+\$53.8	63.6%
Neutral	\$26.1	\$45.5	+\$19.4	62.3%
Extreme Fear	\$37.4	\$61.1	+\$23.7	67.8%
Greed	\$43.1	\$27.2	-\$15.9	41.9%
Extreme Greed	\$32.9	\$13.4	-\$19.6	48.8%

Behavioural bias detected

Traders are buying the dip (going long) during Fear even though shorting earns 2.3x more per trade during those same conditions. This is a classic loss-aversion bias — traders anticipate a rebound rather than following the trend downward.

3.3 F&G score bucket analysis

Breaking the 0-100 F&G range into 8 granular buckets reveals a non-monotonic relationship with PnL - the best and worst score zones are not where intuition suggests.

F&G range	Avg PnL (\$)	Trades	Assessment
80–100	\$101.3	14,315	Highest PnL — euphoria creates large moves
30–40	\$56.9	28,639	Fear recovery zone — good entry conditions
0–20	\$52.1	13,139	Extreme panic — contrarians rewarded
40–50	\$51.4	42,333	Neutral-fear boundary — moderate returns
70–80	\$50.1	50,641	High greed — largest trade volume bucket
50–60	\$45.2	15,702	Mild greed — diminishing returns begin
20–30	\$22.6	30,787	Deep fear — traders hesitant, low PnL
60–70	\$23.5	15,662	Dead zone — weakest PnL in greed territory

4. Trader-Level Analysis

4.1 Overall performance ranking

The cohort spans a wide performance range - from \$2.1M net profit to -\$169K net loss. Fee drag is minimal for most top performers but becomes critical for high-frequency traders.

Trader (short)	Total PnL	Net (after fees)	Win rate	Trades	Archetype
0x0833...9012	\$1,600K	\$1,593K	36.0%	3,818	Short Specialist
0xbaaa...7864	\$940K	\$932K	46.8%	21,192	Long Biased
0xb123...ed23	\$725K	\$2,127K*	34.3%	11,945	Balanced
0x4acb...b9f4	\$685K	\$670K	48.9%	4,332	Long Biased
0x513b...4ff1	\$646K	\$764K	39.5%	11,257	Long Biased
0x75f7...70d4	\$379K	\$377K	81.2%	9,850	Market Maker
0x8170...a63b	-\$156K	-\$169K	38.4%	4,572	Short Specialist
0xb899...2115	\$22K	-\$6K	43.9%	4,838	Fear Trader

* 0xb123 total PnL from position type subset; full account total is \$2.14M.

4.2 Trader archetypes

Five distinct archetypes emerge from classifying traders on win rate, long/short ratio, and average F&G score when active:

Archetype	Classification criteria	Profitable examples	Characteristic
Short Specialist	Long ratio < 30%	0x0833, 0x72c6	Profits from shorting; fails on trend reversal
Long Biased	Long ratio > 70%	0xbaaa, 0x4acb	Directional momentum; exposed to drawdowns
Market Maker	Win rate > 70%	0x75f7	Liquidity provision; spread capture; not directional
Fear Trader	Avg F&G < 40	0xbee1	Most active in low-sentiment environments
Greed Trader	Avg F&G > 60	0x92f1	Most active in high-sentiment environments
Balanced	No dominant signal	0xb123, 0x7274	Adaptive; trades both sides across regimes

4.3 The mirror traders

Two of the top-5 traders by total PnL operate completely opposing strategies, yet both are highly profitable. This is one of the most instructive findings in the dataset.

	0x0833 (Short Specialist)	0xb123 (Balanced)
Total PnL	\$1.60M	\$2.14M
Best regime	Extreme Fear (\$1,248 avg)	Extreme Greed (\$673 avg)
Worst regime	Extreme Greed (-\$43 avg)	Extreme Fear (\$13 avg)
Direction	100% Short	Mixed (38% Long, 62% Short)
Mechanism	Short into panic selling	Short into euphoria peaks

Strategic implication

A sentiment-conditional strategy that switches from 0x0833-style shorting during Fear to 0xb123-style positioning during Greed could theoretically capture the best of both. The data supports sentiment-regime-aware position sizing.

4.4 The market maker (0x75f7)

Trader 0x75f7 achieves an 81.2% win rate across 9,850 trades — the highest in the cohort — and \$379K total PnL. Their direction mix is the key: 4,381 Close Short and 4,256 Close Long positions, vs only 667 Open Short and 535 Open Long. This profile is consistent with a liquidity provision or statistical arbitrage strategy, not directional prediction.

- Win rate peaks at 91.5% during Extreme Fear (316 trades) and dips to 71.5% in Neutral — spread compression in low-volatility environments reduces edge.
- Average PnL goes slightly negative (-\$1.78) in Neutral, the only regime where this trader loses money on average.
- Top coins are JELLY (99 win rate), OGN (99.6% on 258 trades), GRIFFAIN (93%), and PURR (91.3%) — small-cap coins with wider spreads.

Warning

This trader's 81% win rate should not be interpreted as a signal to copy their directional bets. The mechanism is spread capture, not market prediction. Copying position direction without the underlying liquidity infrastructure would not replicate the result.

5. Temporal & Volume Analysis

5.1 Activity explosion in 2025

Trade activity grew dramatically across the observation window, from under 2,500 trades/month in early 2024 to 52,358 trades in April 2025. This expansion was driven by multiple new traders joining the cohort simultaneously in late 2024, not by existing traders scaling up.

- November 2024 marks the inflection point - trade count jumps from ~1,200/month to 6,538, driven by the bull market triggered by the US election results (F&G avg 83 in November 2024).
- Despite a sharp F&G decline in early 2025 (F&G fell to 31 in March 2025), trade count and avg PnL remained elevated. Traders adapted to the fear environment rather than exiting.
- August 2024 was the worst month: avg PnL -\$84.64, F&G avg 34.9. The cohort had not yet developed Fear-regime strategies.

5.2 Sentiment momentum as a signal

Beyond the absolute F&G score, the direction of change in the F&G index provides additional signal:

F&G direction	Avg next-day PnL	Median PnL	Days
Stable (change < 5)	\$70.43	\$19.56	317
Falling	\$48.66	\$18.30	86
Rising	\$29.09	\$14.10	73

Stable sentiment days produce 2.4x the avg PnL of rising-sentiment days. This suggests that transitions - when sentiment is actively shifting upward - are the hardest regime to trade, likely because the market is in price discovery and signals are mixed.

6. Coin Concentration & Fee Analysis

6.1 Top coins by PnL

Coin	Trades	Avg PnL (\$)	Total PnL (\$)	Assessment
@107	29,992	\$92.8	\$2.78M	Highest total PnL; likely an index perpetual

Coin	Trades	Avg PnL (\$)	Total PnL (\$)	Assessment
HYPE	68,005	\$28.7	\$1.95M	Highest volume; solid per-trade return
SOL	10,691	\$153.4	\$1.64M	Best avg PnL per trade among major coins
ETH	11,158	\$118.3	\$1.32M	High value, high conviction trades
BTC	26,064	\$33.3	\$868K	Large volume, moderate per-trade return
TRUMP	1,920	-\$190	-\$365K	Catastrophic loss leader; one trader -\$433K
FARTCOIN	4,650	-\$21.7	-\$101K	Negative aggregate despite some winners

6.2 Fee drag

Fees are low relative to PnL for most top performers, but become a critical issue for high-frequency traders with marginal edge.

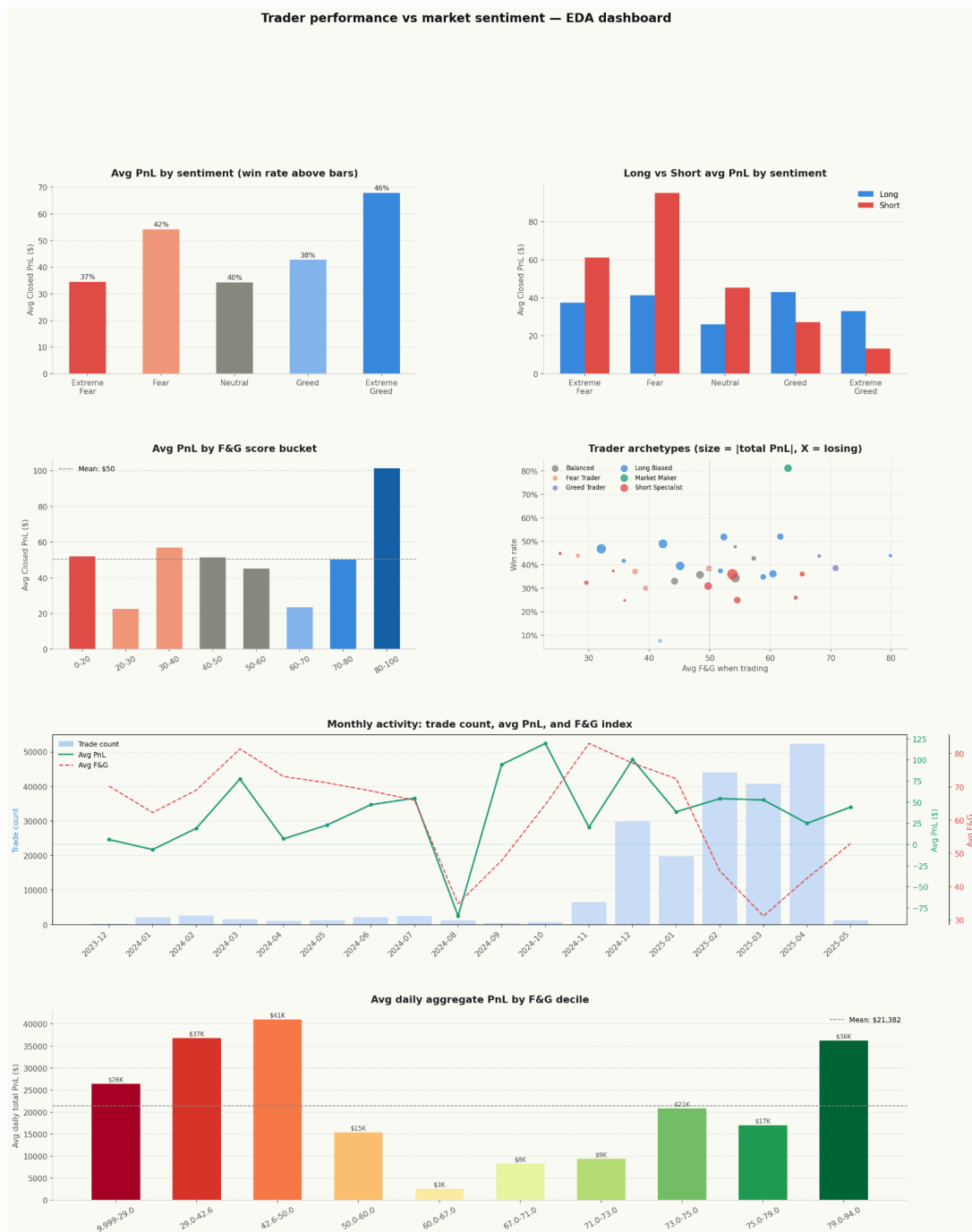
- 0x0833: \$7.4K fees on \$1.6M PnL - fee drag of 0.46%. Exceptional efficiency.
- 0x513b: \$76.4K fees on \$840K PnL - 9.1% fee drag. High-frequency strategy with diminishing edge.
- 0xb899: \$28.6K fees on \$22.5K gross PnL - net negative at -\$6.2K. Trading itself into insolvency via fees.

Rule of thumb

For high-frequency strategies in this cohort, fee drag above 5% of gross PnL is a danger signal. Net PnL (gross minus fees) is the only meaningful performance metric.

7. Summary Dashboard

The following six-panel dashboard consolidates all major findings from Sections 3–6 into a single visual reference.



8. Key Takeaways & Strategy Implications

8.1 Trade with the extremes, not the middle

F&G scores of 0–20 and 80–100 both outperform the 60–67 dead zone. Position sizing should increase during extreme regimes and decrease during mild greed transitions.

8.2 Short more during fear

During Fear and Extreme Fear, short positions earn 2-2.3x the PnL of long positions yet only 32-37% of trades are shorts. The data supports a regime-conditional bias toward shorting during fear environments.

8.3 Stable sentiment > shifting sentiment

Days when the F&G index barely moves produce the highest avg PnL (\$70). Avoid overtrading on days when sentiment is actively rising (lowest avg PnL: \$29) - the market is transitioning and signals are noise.

8.4 Sentiment score does not directly predict PnL

The correlation between F&G value and daily total PnL is -0.08 - effectively zero. What matters is the trading activity the sentiment generates (trade count correlates at +0.36). Strategy and execution matter more than reading the sentiment correctly.

8.5 Avoid meme coins during sentiment extremes

TRUMP (-\$365K) and FARTCOIN (-\$101K) demonstrate that high-volatility meme coins during extreme sentiment create asymmetric downside. The winners in these coins are few; the losers are large. Position limits on speculative assets are warranted.

8.6 Monitor fee-to-PnL ratio

Any strategy crossing 5% fee drag on gross PnL should be reviewed. One trader in this cohort is net-negative solely due to fees despite a 43.9% win rate. Net PnL is the only valid performance measure.

9. Findings Summary

#	Finding	Implication
1	Extreme Greed: highest PnL (\$67.9) and win rate (46.5%)	Do not exit positions prematurely during euphoria
2	Shorts outperform longs in 4 of 5 regimes	Rebalance long bias toward shorts during Fear
3	F&G 60–67 is a dead zone (\$2.5K daily PnL)	Reduce sizing during mid-greed transitions
4	Stable sentiment days: 2.4x PnL of rising days	Avoid overtrading on sentiment shift days
5	Two mirror traders — Fear vs Greed specialists — both profitable	Sentiment-conditional strategy switching is viable
6	Market maker achieves 81% win rate via spread capture	High win rate $\neq$ directional edge; check mechanism
7	TRUMP + FARTCOIN: -\$465K combined	Cap meme coin exposure during sentiment extremes
8	0xb899 is fee-insolvent despite positive gross PnL	Net PnL after fees is the only valid metric
9	Activity exploded 20x in late 2024	Cohort is scaling; earlier patterns may not hold at scale
10	F&G has near-zero correlation with PnL ($r = -0.08$)	Sentiment signals execution context, not outcome